

Periodic Second Coefficients through a Prime Target: A Bombieri–Vinogradov Transfer, Arithmetic Involutions, and the Combined-Modulus Boundary

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Abstract

Let $x = MN$, let $h \neq 0$ be fixed, and let $q \geq 3$ be odd with $(q, h) = 1$. For a bounded q -periodic function β supported on the unit classes modulo q , we study the two-scale prime-target form

$$\sum_{M < m \leq 2M} \alpha_m \sum_{n \geq 1} \beta(n) \Lambda(mn + h) W(n/N), \quad |\alpha_m| \leq 1.$$

Allowing an arbitrary bounded second coefficient leads to the unresolved bilinear/Type II interface isolated in the preceding work. In the low-period range

$$q \leq (\log x)^C, \quad 2Mq \leq x^{1/2} (\log x)^{-B},$$

we prove instead an exact coefficient-transfer law, with arbitrary logarithmic saving, by a direct application of the classical maximal Bombieri–Vinogradov theorem. The main term is operator-valued:

$$N \int W \sum_m \alpha_m \frac{m}{\varphi(m)} (\mathbf{P}_{q,h} \beta)(m), \quad (\mathbf{P}_{q,h} \beta)(a) = \frac{1}{\varphi(q)} \sum_{\substack{b \in (\mathbb{Z}/q\mathbb{Z})^\times \\ (ab+h, q)=1}} \beta(b).$$

Thus the periodic coefficient is not replaced by its mean: it is transported by an explicit finite arithmetic operator.

We determine this operator completely. After row normalization it is a self-adjoint Markov operator. Character modes with a conductor divisible by p^2 are annihilated. A surviving character of exact squarefree conductor r is exchanged with its conjugate and multiplied by

$$(-1)^{\omega(r)} \overline{\chi(-h)} \prod_{p|r} (p-2)^{-1}.$$

This is exactly the finite-conductor multiplier found for the centered rough kernel in the preceding work, now transferred through the genuine prime target $\Lambda(mn + h)$. We also give the corresponding prime-weighted source corollary and an exact finite certificate.

The analytic input is classical; no new level of distribution is asserted. The modulus is mq , not m , and residue splitting costs the period q . Moreover, the finite-conductor energy-escape theorem from the preceding paper shows that polylogarithmic periodic spaces capture a vanishing proportion of the L^2 energy of $\Lambda - 1$. Consequently one cannot substitute $\beta = \Lambda - 1$, and the result gives neither a fixed-shift prime-pair asymptotic nor a twin-prime lower bound or a breach of the parity barrier.

Keywords: periodic coefficient; prime target; Bombieri–Vinogradov theorem; arithmetic involution; finite-conductor spectrum; sieve parity.

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1 Introduction and exact scope

The transition from a rough target to a prime target changes the analytic problem. A finite sieve kernel can test the congruences $mn + h \not\equiv 0 \pmod{p}$ for prescribed primes p , but primality is encoded by the von Mangoldt weight $\Lambda(mn + h)$. The preceding paper [14] made this distinction explicit: a one-sided Bombieri–Vinogradov argument controls

$$\sum_{M < m \leq 2M} \alpha_m \sum_{n \geq 1} \Lambda(mn + h) W(n/N), \quad |\alpha_m| \leq 1, \quad (1)$$

in the Type I range $M \leq x^{1/2}(\log x)^{-B}$, where $x = MN$. Inserting an arbitrary bounded coefficient β_n in the n -sum destroys the direct reduction to one residue class modulo m . The resulting form leads to the unresolved bilinear/Type II interface isolated there.

This paper treats a deliberately narrower coefficient class. We suppose that β is periodic modulo a small integer q and is supported on the unit classes. Splitting n modulo q turns each row into finitely many prime sums in reduced residue classes modulo

$$mq. \quad (2)$$

The classical maximal Bombieri–Vinogradov theorem can then be applied after two costs have been recorded: the level condition concerns Mq , and the triangle inequality over the residue classes costs at most q . Both costs are harmless when

$$q \leq (\log x)^C, \quad 2Mq \leq \frac{x^{1/2}}{(\log x)^B}. \quad (3)$$

They are not harmless for a general power-sized period.

The principal point is not merely that a periodic coefficient can be handled. Its main term retains a finite operator. For an odd q with $(q, h) = 1$, put

$$(\mathbf{P}_{q,h}\beta)(a) = \frac{1}{\varphi(q)} \sum_{\substack{b \in G_q \\ (ab+h,q)=1}} \beta(b), \quad G_q = (\mathbb{Z}/q\mathbb{Z})^\times. \quad (4)$$

The transfer theorem proved below has the rowwise main term

$$\frac{mN}{\varphi(m)} \left(\int_0^\infty W(t) dt \right) (\mathbf{P}_{q,h}\beta)(m). \quad (5)$$

Thus the prime target does not average β to a scalar. It transports the finite residue observable by $\mathbf{P}_{q,h}$.

1.1 Results and logical status

The paper contains four groups of statements.

- (i) We determine $\mathbf{P}_{q,h}$ exactly. After division by its row density

$$\kappa_q = \prod_{p|q} \frac{p-2}{p-1}, \quad (6)$$

it is a self-adjoint Markov operator. If a local character has conductor exponent at least two at some $p \mid q$, that mode is annihilated. Every surviving nonprincipal mode has squarefree exact conductor r and is exchanged with its conjugate with modulus

$$d(r) = \prod_{p \mid r} \frac{1}{p-2}. \quad (7)$$

The phase is the arithmetic involution phase $(-1)^{\omega(r)} \overline{\chi(-h)}$. This is the same exact multiplier that occurs in the rough centered kernel of Wang [12]. The structural contribution here is to identify and prove the occurrence of this multiplier in the main term of the two-scale prime-target form.

- (ii) We prove an ℓ^1 row-transfer theorem in the range (3). It is uniform over all bounded unit-supported q -periodic coefficients and immediately gives coefficient-uniform bilinear forms. The only analytic input is the classical maximal Bombieri–Vinogradov theorem [1, 7, 11].
- (iii) We give exact character corollaries and add a Mangoldt weight in the m -coordinate. Siegel–Walfisz then removes every nonprincipal periodic output mode. This produces a two-prime/periodic-factor asymptotic in a strongly unbalanced Type I range. It does not make the periodic factor prime.
- (iv) We combine the transfer theorem with the finite-conductor energy-escape result of Wang [13]. Periods of size $(\log x)^C$ capture a vanishing proportion of the L^2 energy of $\Lambda - 1$. Therefore the present theorem cannot be specialized to the missing prime coefficient. We record this as a strict theorem-interface statement, not as a universal impossibility result.

The finite computations have the same separation. An integer/rational certificate verifies the local identities, conductor multipliers, involutions, and exact arithmetic-progression reindexing. A separate floating-point experiment displays finite Mangoldt rows. No analytic theorem depends on either computation.

1.2 Relation to known distribution results

The estimate in this paper is an organized consequence of the standard square-root level of distribution. It is not a new Bombieri–Vinogradov theorem. Results beyond the square-root level are known for special well-factorable weights and fixed residue structures [2, 10]; those hypotheses do not directly cover the residue $mb + h$ modulo the moving combined modulus mq with an arbitrary periodic coefficient.

Much deeper cancellation estimates are known for isolated shifted-prime character sums, beginning with earlier shifted-prime work and continuing through stronger bounds of Kerr, as well as for general trace functions over primes [4, 6, 8]. Our object and purpose are different: the source scale m moves, the modulus is mq , arbitrary row coefficients are retained, and the desired output is the complete finite operator $P_{q,h}$. We make no priority claim for a new estimate of an isolated character sum.

An L^2 treatment using Barban–Davenport–Halberstam or variational forms [9] may reduce the elementary residue-splitting loss in some ranges. In the small-modulus regime, however, standard formulations also contain terms that do not automatically yield the power-sized-period version of our coefficient-uniform statement. Such an extension is left as an open analytic problem rather than inferred from an inapplicable mean square bound.

1.3 What is not proved

For clarity, none of the following is a consequence of this paper.

- There is no estimate with an arbitrary bounded second coefficient.
- There is no bound for the Vaughan Type II hard packet identified in Wang [14].
- There is no asymptotic for $\sum_n \Lambda(n)\Lambda(n+2)$ and no twin-prime lower bound.
- The parity obstruction is not broken. The relevant limitations of finite sieve information remain those discussed in the asymptotic-sieve and parity literature [3, 5].
- The finite Markov spectrum below is not a spectral realization of the primes. It is the exact Fourier form of a local congruence operator.

The gain is narrower but rigorous: one nontrivial second-coefficient class crosses the prime target at a prescribed shift, and the surviving arithmetic dynamics is computed exactly.

2 The finite prime-target operator

Throughout this section $q \geq 3$ is odd, $(q, h) = 1$, and $G_q = (\mathbb{Z}/q\mathbb{Z})^\times$. We use normalized counting measure on G_q . For $\beta : G_q \rightarrow \mathbb{C}$, define

$$(\mathbf{P}_{q,h}\beta)(a) = \frac{1}{\varphi(q)} \sum_{\substack{b \in G_q \\ (ab+h,q)=1}} \beta(b), \quad (8)$$

$$\kappa_q = \prod_{p|q} \frac{p-2}{p-1}, \quad \mathbf{T}_{q,h} = \kappa_q^{-1} \mathbf{P}_{q,h}. \quad (9)$$

For each a , exactly

$$R_q = \varphi(q)\kappa_q = \prod_{p^e \parallel q} p^{e-1}(p-2) \quad (10)$$

values of b are admissible. Consequently $\mathbf{T}_{q,h}$ is a Markov operator. Its kernel is real and symmetric, because $ab+h=ba+h$, so it is self-adjoint on $L^2(G_q)$.

2.1 Inclusion-exclusion and the arithmetic involutions

For every squarefree $d \mid \text{rad}(q)$ let

$$J_{h,d}(a) = -ha^{-1} \pmod{d}, \quad a \in G_d, \quad (11)$$

and let $U_{h,d}f = f \circ J_{h,d}$. Then $J_{h,d}^2$ is the identity, so $U_{h,d}$ is a self-adjoint unitary involution. Define the normalized fiber average

$$(E_{q,d}\beta)(u) = \frac{\varphi(d)}{\varphi(q)} \sum_{\substack{b \in G_q \\ b \equiv u \pmod{d}}} \beta(b). \quad (12)$$

Proposition 2.1 (Exact involution bundle). *For every $\beta : G_q \rightarrow \mathbb{C}$ and $a \in G_q$,*

$$(\mathbf{P}_{q,h}\beta)(a) = \sum_{d \mid \text{rad}(q)} \frac{\mu(d)}{\varphi(d)} (U_{h,d}E_{q,d}\beta)(a \bmod d). \quad (13)$$

All terms in this identity are finite and exact.

Proof. Inclusion-exclusion gives

$$\mathbf{1}_{(ab+h,q)=1} = \sum_{\substack{d \mid \text{rad}(q) \\ ab \equiv -h \pmod{d}}} \mu(d).$$

For fixed a and d , the congruence selects the fiber $b \equiv J_{h,d}(a) \pmod{d}$. By (12), its unnormalized sum is

$$\frac{\varphi(q)}{\varphi(d)}(E_{q,d}\beta)(J_{h,d}(a)).$$

Divide by $\varphi(q)$ and sum over d . □

This formula identifies the exact link with the target-divisor involutions of Wang [14]. It is not a mixing assertion: each $U_{h,d}$ is an involution, while the damping comes from the normalized combination of all divisor layers and from the fiber averages.

2.2 Complete character-block decomposition

Let $\widehat{G}_q$ be the character group. We use the Fourier convention

$$B(\chi) = \frac{1}{\varphi(q)} \sum_{b \in G_q} \beta(b) \chi(b), \quad \beta(b) = \sum_{\chi \in \widehat{G}_q} B(\chi) \overline{\chi(b)}. \quad (14)$$

Write $q = \prod_{p \mid q} p^e$. Let $f_p(\chi)$ be the conductor exponent of the local component χ_p modulo p^e .

Call χ *surviving* if $f_p(\chi) \leq 1$ for every $p \mid q$. For such a character put

$$r_\chi = \prod_{\substack{p \mid q \\ f_p(\chi)=1}} p, \quad d(r_\chi) = \prod_{p \mid r_\chi} \frac{1}{p-2}, \quad (15)$$

and

$$c_h(\chi) = (-1)^{\omega(r_\chi)} d(r_\chi) \overline{\chi(-h)}. \quad (16)$$

The principal character is surviving, with $r_\chi = 1$ and $c_h(\chi) = 1$.

Theorem 2.2 (Exact local spectrum). *For every $\beta : G_q \rightarrow \mathbb{C}$,*

$$(\mathbf{P}_{q,h}\beta)(a) = \kappa_q \sum_{\substack{\chi \in \widehat{G}_q \\ \chi \text{ surviving}}} B(\chi) c_h(\chi) \chi(a). \quad (17)$$

Equivalently,

$$\mathbf{T}_{q,h}\overline{\chi} = \begin{cases} c_h(\chi) \chi, & \chi \text{ surviving,} \\ 0, & \chi \text{ not surviving.} \end{cases} \quad (18)$$

In particular, on a surviving mode,

$$\mathbf{T}_{q,h}^2 \chi = d(r_\chi)^2 \chi. \quad (19)$$

The nonzero spectrum consists of 1, the signed eigenvalues on real characters, and the pairs $\pm d(r_\chi)$ on nonreal conjugate pairs. Zero has multiplicity

$$\varphi(q) - \varphi(\text{rad}(q)). \quad (20)$$

For every integer $k \geq 1$,

$$\text{tr}(\mathbf{T}_{q,h}^{2k}) = \prod_{p \mid q} (1 + (p-2)^{1-2k}). \quad (21)$$

Proof. It is enough to compute locally. Fix $p^e \parallel q$, write a for a unit modulo p^e , and consider the sum of $\overline{\chi_p(b)}$ over unit b with $ab + h \not\equiv 0 \pmod{p}$.

If $f_p = 0$, the summand is 1 and the sum is $p^{e-1}(p-2)$. If $f_p = 1$, the complete unit sum is zero and the excluded fiber has p^{e-1} points, on all of which the character has the same value. The allowed sum is therefore

$$-p^{e-1}\overline{\chi_p(-h)}\chi_p(a).$$

If $f_p \geq 2$, both the complete unit sum and the sum over the excluded fiber vanish: the restriction of χ_p to the kernel of reduction modulo p is nonprincipal. Hence this local factor is zero.

Multiplying the local sums and dividing by $\varphi(q)$ proves (17) and (18). Since $c_h(\overline{\chi}) = \overline{c_h(\chi)}$ and $|c_h(\chi)| = d(r_\chi)$, applying $\mathsf{T}_{q,h}$ twice proves (19). There are $p-1$ local characters with conductor exponent at most one, so the nonzero subspace has dimension $\prod_{p|q}(p-1) = \varphi(\text{rad}(q))$. Finally, group the surviving characters by exact conductor. At a prime p , there is one principal choice and $p-2$ nonprincipal choices. Thus

$$\sum_{\chi \text{ surviving}} d(r_\chi)^{2k} = \prod_{p|q} \left(1 + (p-2)(p-2)^{-2k}\right),$$

which is (21). □

Corollary 2.3 (Gap and the prime 3 obstruction). *Suppose q is squarefree. If $3 \mid q$, a nonprincipal conductor-3 mode has modulus-one eigenvalue, so there is no spectral gap on the mean-zero subspace. If $3 \nmid q$ and $p_{\min}$ is the least prime dividing q , then*

$$\|\mathsf{T}_{q,h}\|_{\mathbf{1}^\perp \rightarrow \mathbf{1}^\perp} = \frac{1}{p_{\min} - 2} < 1. \quad (22)$$

For nonsquarefree q , the same norm formula holds on $\mathbf{1}^\perp$; equivalently, it holds on the mean-zero part of the surviving subspace. The orthogonal complement of the surviving subspace is killed in one step.

The finite chain can therefore have a genuine local gap. This does not say that the primes, or the sequence $\Lambda(mn+h)$, form a mixing dynamical system. The distinction will be maintained throughout.

3 A periodic coefficient through the prime target

Fix a nonzero integer h and a function $W \in C_c^1((0, \infty))$. Write

$$\mathcal{I}_W = \int_0^\infty W(t) dt. \quad (23)$$

All constants below may depend on h and W . The parameter h is fixed; in particular, $h = 2$ is permitted. The assertion is nevertheless a theorem about a periodic coefficient in a prime-target form, not a prime-pair theorem.

Let $x = MN$, where $M, N \geq 2$. For an odd integer q coprime to h , let $\beta : G_q \rightarrow \mathbb{C}$, extend it periodically to the integers, and set it equal to zero on the nonunit residue classes. For $(m, hq) = 1$, define the row

$$\mathcal{R}_{m,h}^{(N)}(\beta) = \sum_{n \geq 1} \beta(n) \Lambda(mn+h) W(n/N). \quad (24)$$

3.1 The rowwise transfer theorem

We first state the classical input in the form used in the proof. Put

$$\Delta(Y; r) = \max_{\substack{a \bmod r \\ (a, r) = 1}} \max_{y \leq Y} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod{r}}} \Lambda(n) - \frac{y}{\varphi(r)} \right|. \quad (25)$$

For every $L > 0$, the maximal Bombieri-Vinogradov theorem gives a $B_0 = B_0(L) > 0$ such that

$$\sum_{r \leq Y^{1/2} (\log Y)^{-B_0}} \Delta(Y; r) \ll_L \frac{Y}{(\log Y)^L}. \quad (26)$$

This is the classical theorem [1, 7, 11], not a result proved here.

Theorem 3.1 (Operator-valued periodic transfer). *Fix $A, C > 0$. There is $B = B(A, C, h, W) > 0$ such that the following holds for all sufficiently large $x = MN$. Suppose*

$$3 \leq q \leq (\log x)^C, \quad q \text{ odd}, \quad (q, h) = 1, \quad (27)$$

and

$$2Mq \leq \frac{x^{1/2}}{(\log x)^B}. \quad (28)$$

Uniformly for every $\beta : G_q \rightarrow \mathbb{C}$ with $\|\beta\|_\infty \leq 1$,

$$\sum_{\substack{M < m \leq 2M \\ (m, hq) = 1}} \left| \mathcal{R}_{m, h}^{(N)}(\beta) - \frac{mN}{\varphi(m)} \mathcal{I}_W(\mathbf{P}_{q, h} \beta)(m \bmod q) \right| \ll_{A, C, h, W} \frac{x}{(\log x)^A}. \quad (29)$$

Consequently, for arbitrary complex coefficients $|\alpha_m| \leq 1$ supported on the same rows,

$$\begin{aligned} & \sum_{M < m \leq 2M} \alpha_m \sum_{n \geq 1} \beta(n) \Lambda(mn + h) W(n/N) \\ &= N \mathcal{I}_W \sum_{M < m \leq 2M} \alpha_m \frac{m}{\varphi(m)} (\mathbf{P}_{q, h} \beta)(m) + O_{A, C, h, W} \left(\frac{x}{(\log x)^A} \right). \end{aligned} \quad (30)$$

The theorem is stronger than a single bilinear estimate: it controls the sum of the absolute row errors. This is exactly the supremum of the error in (30) over all row coefficients of modulus at most one.

Proof. Put

$$\Lambda_q(t) = \Lambda(t) \mathbf{1}_{(t, q) = 1}. \quad (31)$$

We first prove the result with Λ_q in place of Λ . Split the n -sum into $b \in G_q$. If $(mb + h, q) = 1$, the substitution $t = mn + h$ gives the exact correspondence

$$\{n \geq 1 : n \equiv b \pmod{q}\} \longleftrightarrow \{t \in \mathbb{Z} : t \equiv mb + h \pmod{mq}, (t - h)/m \geq 1\}. \quad (32)$$

The class on the right is reduced. Indeed, $(mb + h, m) = (h, m) = 1$ and it is coprime to q by assumption. On this class $\Lambda_q(t) = \Lambda(t)$ whenever $t \geq 1$. Equivalently, after inserting the transformed weight $W((t - h)/(mN))$, the target sum is a weighted prime sum in that reduced class modulo mq .

Stieltjes partial summation, applied to $W((t - h)/(mN))$, gives

$$\sum_{\substack{n \geq 1 \\ n \equiv b \pmod{q}}} \Lambda_q(mn + h) W(n/N)$$

$$= \frac{mN}{\varphi(mq)} \mathcal{I}_W + E_{m,b}, \quad (33)$$

where, for a constant $C_W > 0$,

$$|E_{m,b}| \ll_W \Delta(C_W x; mq). \quad (34)$$

For sufficiently large x , the support of the transformed weight lies in the positive t -axis. The continuous main term is exactly

$$\frac{1}{\varphi(mq)} \int_0^\infty W((t-h)/(mN)) dt = \frac{mN}{\varphi(mq)} \mathcal{I}_W.$$

If $(mb+h, q) > 1$, the corresponding Λ_q sum is identically zero.

Because $(m, q) = 1$, we have $\varphi(mq) = \varphi(m)\varphi(q)$. Summing the main terms in (33) against $\beta(b)$ therefore gives

$$\frac{mN}{\varphi(m)\varphi(q)} \mathcal{I}_W \sum_{\substack{b \in G_q \\ (mb+h, q)=1}} \beta(b) = \frac{mN}{\varphi(m)} \mathcal{I}_W (\mathbf{P}_{q,h,\beta})(m).$$

For the errors, the triangle inequality and (34) give

$$\begin{aligned} & \sum_{\substack{M < m \leq 2M \\ (m,hq)=1}} \left| \sum_{\substack{b \in G_q \\ (mb+h, q)=1}} \beta(b) E_{m,b} \right| \\ & \ll_W \left(\sum_{b \in G_q} |\beta(b)| \right) \sum_{r \leq 2Mq} \Delta(C_W x; r). \end{aligned} \quad (35)$$

Here we enlarged the moduli mq to all $r \leq 2Mq$. Choose the exponent in (26) larger than $A + C + 2$ and then choose B in (28) accordingly. Since

$$\sum_{b \in G_q} |\beta(b)| \leq \varphi(q) \leq q \leq (\log x)^C, \quad (36)$$

the right side of (35) is $O_{A,C,h,W}(x(\log x)^{-A})$.

It remains to restore the terms deleted in (31). If $(mn+h, q) > 1$ and $\Lambda(mn+h) \neq 0$, then

$$mn+h = p^k \quad (37)$$

for some prime $p \mid q$. On the support of the two scales, $mn \asymp_W x$. For each p , only $O_W(1)$ powers p^k lie in the resulting fixed-ratio interval. For each such power, the number of pairs (m, n) is at most $\tau(|p^k - h|)$. Hence, for every fixed $\eta > 0$, the total deleted contribution, even after absolute values over all rows, is

$$\ll_{\eta,h,W} x^\eta \sum_{p \mid q} \log p \ll_{\eta,h,W} x^\eta \log(2q). \quad (38)$$

The divisor bound $\tau(t) \ll_\eta t^\eta$ was used here. Under (27), (38) is absorbed by $x(\log x)^{-A}$. This proves (29); multiplying the row errors by α_m proves (30). $\square$

Remark 3.2 (The unsimplified complexity bound). Before imposing $q \leq (\log x)^C$, the proof gives, for every $L > 0$ and in the corresponding Bombieri–Vinogradov modulus range,

$$\sum_m |\text{row error}_m| \ll_{L,h,W} \|\beta\|_{\ell^1(G_q)} \frac{x}{(\log x)^L} + O_{\eta,h,W}(x^\eta \log(2q)). \quad (39)$$

This formula explains the polylogarithmic condition. The classical theorem controls the maximum of one reduced class for each modulus; summing an arbitrary observable over all q classes is not free.

Remark 3.3 (The combined-modulus boundary). The modulus in (32) is mq . It would be incorrect to retain the one-sided condition $M \leq x^{1/2}(\log x)^{-B}$ from (1) after introducing a period q . The valid direct condition is (28). Likewise, a character of small exact conductor inflated to a much larger ambient period still contains the unit cutoff for that ambient period; its integer-period cost is the ambient q , unless the coefficient is rewritten at its genuinely smaller period.

4 Modal transfer and a prime-weighted source

The operator theorem becomes especially transparent in character coordinates. Continue to use the Fourier convention (14). Combining theorems 2.2 and 3.1 gives the exact main term

$$\begin{aligned} & \sum_m \alpha_m \sum_n \beta(n) \Lambda(mn + h) W(n/N) \\ &= \kappa_q N \mathcal{I}_W \sum_m \alpha_m \frac{m}{\varphi(m)} \sum_{\substack{\chi \in \widehat{G}_q \\ \chi \text{ surviving}}} B(\chi) c_h(\chi) \chi(m) + O_{A,C,h,W} \left(\frac{x}{(\log x)^A} \right). \end{aligned} \quad (40)$$

The sums over m have the support and the range of theorem 3.1.

4.1 One character mode

Corollary 4.1 (Primitive squarefree mode). *Let q be odd and squarefree, let χ be primitive modulo q , and suppose the hypotheses of theorem 3.1 hold. Then*

$$\begin{aligned} & \sum_m \alpha_m \sum_{n \geq 1} \overline{\chi(n)} \Lambda(mn + h) W(n/N) \\ &= \frac{\mu(q) \overline{\chi(-h)}}{\varphi(q)} N \mathcal{I}_W \sum_m \alpha_m \frac{m}{\varphi(m)} \chi(m) + O_{A,C,h,W} \left(\frac{x}{(\log x)^A} \right). \end{aligned} \quad (41)$$

Here every Dirichlet character is extended by zero away from the unit classes. Replacing $\overline{\chi}$ by χ conjugates the phase and the output character.

Proof. For a primitive character of squarefree modulus q , $r_\chi = q$ and

$$\kappa_q c_h(\chi) = \prod_{p|q} \frac{p-2}{p-1} \frac{\mu(q) \overline{\chi(-h)}}{\prod_{p|q} (p-2)} = \frac{\mu(q) \overline{\chi(-h)}}{\varphi(q)}.$$

Now apply (40). □

This formula displays the arithmetic involution directly:

$$\overline{\chi}(n) \longmapsto \overline{\chi(-h)} \chi(m). \quad (42)$$

The scalar $\mu(q)/\varphi(q)$ is not a heuristic independence factor. It is the exact local main coefficient after the q residue classes have been passed through the prime target.

Corollary 4.2 (Higher conductor-exponent annihilation). *Let $\chi \in \widehat{G}_q$ have local conductor exponent at least two at some $p | q$. Under the hypotheses of theorem 3.1,*

$$\sum_m \alpha_m \sum_{n \geq 1} \overline{\chi(n)} \Lambda(mn + h) W(n/N) \ll_{A,C,h,W} \frac{x}{(\log x)^A}. \quad (43)$$

The vanishing is a local algebraic cancellation in the leading term. It is not a zero-free-region statement about the L -function of χ .

4.2 Exact-conductor truncations

For $1 \leq R \leq \text{rad}(q)$, define the surviving low-conductor part

$$\beta_{\leq R}(b) = \sum_{\substack{\chi \text{ surviving} \\ r_\chi \leq R}} B(\chi) \overline{\chi(b)}. \quad (44)$$

Its transferred main term is obtained from (40) by restricting the same sum to $r_\chi \leq R$. There is, however, a distinction between exact conductor and actual integer period.

If (44) is retained as a function on G_q and extended by zero off the units modulo q , its period remains q because of the ambient unit cutoff. The Bombieri–Vinogradov condition is then on Mq . If one instead starts with a primitive character of its genuine modulus r , as in [corollary 4.1](#), the condition is on Mr . This prevents a low exact-conductor label inside an enormous ambient CRT space from being used to evade the combined-modulus boundary.

For a finite sum of primitive modes of possibly different genuine periods, the proof applies term by term. A sufficient polylogarithmic complexity budget, with every r_j odd and $(r_j, h) = 1$, is

$$\sum_j |\gamma_j| r_j \leq (\log x)^C, \quad 2M \max_j r_j \leq x^{1/2} (\log x)^{-B}. \quad (45)$$

This is a convenient corollary of the residue-class proof, not an optimal large-sieve statement.

4.3 Adding a prime weight in the source coordinate

The nonprincipal output modes in (40) need not vanish for arbitrary α_m : one may choose α_m to align with a character. They do vanish after placing a prime weight on the m -coordinate.

Corollary 4.3 (Prime source and periodic factor). *Fix $\delta, A, C > 0$, let $W_1, W_2 \in C_c^1((0, \infty))$, and suppose $\text{supp } W_1 \subset [1, 2]$. Let $q \geq 3$ be odd with $(q, h) = 1$, let $\beta : G_q \rightarrow \mathbb{C}$ satisfy $\|\beta\|_\infty \leq 1$, and extend β q -periodically to the integers and by zero on nonunit residue classes. Put $x = MN$ and assume*

$$M \geq x^\delta, \quad q \leq (\log M)^C, \quad 2Mq \leq \frac{x^{1/2}}{(\log x)^B}, \quad (46)$$

where $B = B(A, C, \delta, h, W_1, W_2)$ is sufficiently large. Then

$$\begin{aligned} & \sum_{\substack{m, n \geq 1 \\ (m, hq) = 1}} \Lambda(m) W_1(m/M) \beta(n) \Lambda(mn + h) W_2(n/N) \\ &= \kappa_q B(\mathbf{1}) MN \left(\int_0^\infty W_1(u) du \right) \left(\int_0^\infty W_2(v) dv \right) + O_{A, C, \delta, h, W_1, W_2} \left(\frac{x}{(\log x)^A} \right), \end{aligned} \quad (47)$$

where

$$B(\mathbf{1}) = \frac{1}{\varphi(q)} \sum_{b \in G_q} \beta(b). \quad (48)$$

In particular, a unit-mean-zero periodic factor gives an arbitrary logarithmic saving.

Proof. Apply [theorem 3.1](#) with row coefficient $\Lambda(m) W_1(m/M)$. Dividing this coefficient by $O(\log x)$ first and requesting one additional logarithmic saving justifies the application. Substitute (17) in the main term.

For the principal character, the prime number theorem and partial summation give

$$\sum_m \Lambda(m) W_1(m/M) \frac{m}{\varphi(m)} = M \int_0^\infty W_1(u) du + O_A \left(\frac{M}{(\log M)^A} \right). \quad (49)$$

The factor $m/\varphi(m)$ may be replaced by 1: on primes the difference contributes $O_{W_1}(\log M)$, and higher prime powers contribute $O_{W_1}(M^{1/2}(\log M)^2)$.

For every nonprincipal $\chi \in \widehat{G}_q$, the Siegel–Walfisz theorem, uniformly for $q \leq (\log M)^C$, gives

$$\sum_m \Lambda(m) W_1(m/M) \frac{m}{\varphi(m)} \chi(m) \ll_{A,C,W_1} \frac{M}{(\log M)^A}. \quad (50)$$

The rows inherit the condition $(m, hq) = 1$. Its excluded prime powers obey the uniform bound

$$\sum_{\substack{M < p^k \leq 2M \\ p|hq}} \Lambda(p^k) \frac{p}{p-1} \ll_h \log(2q), \quad (51)$$

because a fixed prime contributes only $O(1)$ powers to a dyadic interval. They are therefore absorbed in the same saving. There are at most $\varphi(q) \leq (\log M)^C$ surviving modes, while $|B(\chi)|, |c_h(\chi)| \leq 1$. Increasing the Siegel–Walfisz saving absorbs their total. Finally $M \geq x^\delta$ makes $\log M$ comparable to $\log x$, proving (47). $\square$

The range (46) is strongly unbalanced. Ignoring logarithmic factors, the combined-modulus condition says

$$N \gtrsim Mq^2. \quad (52)$$

This is still a Type I theorem. The periodic factor has crossed the prime target, but it has not become a second unrestricted prime weight.

5 The finite-conductor boundary

The transfer theorem is uniform over a nontrivial coefficient class, but that class is small compared with the prime fluctuation. This section makes the missing step quantitative.

For an interval

$$I_N = (N, 2N] \cap \mathbb{Z}, \quad (53)$$

let $V_q(N) \subset \mathbb{C}^{I_N}$ be the restrictions of all q -periodic functions, and let $P_{q,N}$ be orthogonal projection onto $V_q(N)$ in counting-measure $\ell^2(I_N)$. Put

$$g_N(n) = \Lambda(n) - 1. \quad (54)$$

The preceding paper proves the following estimate [13].

Theorem 5.1 (Finite-conductor energy escape). *For every fixed $\delta > 0$, uniformly for $q \leq N^{1-\delta}$,*

$$\|P_{q,N} g_N\|_2^2 \ll_\delta N \frac{q}{\varphi(q)}. \quad (55)$$

Moreover,

$$\|g_N\|_2^2 \sim N \log N. \quad (56)$$

Consequently,

$$\inf_{v \in V_q(N)} \|g_N - v\|_2^2 = \|g_N\|_2^2 - \|P_{q,N} g_N\|_2^2, \quad (57)$$

and hence

$$\inf_{v \in V_q(N)} \|g_N - v\|_2^2 = N \log N + O_\delta \left(N \frac{q}{\varphi(q)} \right) + o(N \log N). \quad (58)$$

For $q \leq (\log N)^C$, the standard estimate $q/\varphi(q) \ll \log \log(3q)$ gives

$$\frac{\|P_{q,N} g_N\|_2^2}{\|g_N\|_2^2} \ll_C \frac{\log \log \log N}{\log N} = o(1). \quad (59)$$

The precise triple logarithm is not important; the vanishing ratio is.

5.1 Why the prime coefficient cannot be inserted

Let $v_q = P_{q,N}g_N$ and $\rho_q = g_N - v_q$. This is the best linear q -periodic approximation in $\ell^2(I_N)$. The present theorem can transport a bounded unit-supported periodic coefficient of period $q \leq (\log x)^C$. Even if one granted a separate extension of the transfer theorem covering v_q , (58) shows that

$$\|\rho_q\|_2^2 = (1 - o(1))N \log N. \quad (60)$$

Thus the untransferred residual contains asymptotically all of the linear prime-fluctuation energy.

Proposition 5.2 (Strict non-specialization). *Neither [theorem 3.1](#) nor [corollary 4.3](#) can be specialized, by periodic projection alone, to $\beta(n) = \Lambda(n) - 1$. Doing so would require a new estimate for the residual ρ_q in (60). The finite-conductor transfer supplies no such estimate.*

Proof. The coefficient $\Lambda - 1$ is neither bounded nor periodic. More decisively, (58) holds after optimizing over the entire linear space of q -periodic functions, which is larger than the unit-supported bounded class of [theorem 3.1](#). Hence no choice in that class makes the ℓ^2 residual negligible. The transfer theorem contains no bilinear bound for the residual, so replacing it by zero would be an additional unproved assertion. $\square$

This is a boundary of the present method and coefficient class. It is not a proof that every nonlocal, nonlinear, dispersion, or Type II construction must fail. In particular, an L^2 energy statement alone does not determine the sign or cancellation of the residual prime-target form.

5.2 The scale inherited from the rough prime window

TPC-5 uses the squarefree rough modulus

$$Q(w) = \prod_{w < p \leq w^2} p, \quad \log Q(w) = (1 + o(1))w^2. \quad (61)$$

To transfer an entire $Q(w)$ -periodic observable through the present theorem, one needs $Q(w) \leq (\log x)^C$. Therefore

$$w^2 \leq (C + o(1)) \log \log x, \quad w = O(\sqrt{\log \log x}). \quad (62)$$

The local prime window may grow, but only to this polylogarithmic-conductor scale under the direct Bombieri–Vinogradov proof. Low exact-conductor modes inside a larger $Q(w)$ do not change its ambient unit cutoff, as noted in [remark 3.3](#).

5.3 A precise next-step ledger

The current interface can be summarized as

$$\underbrace{\text{periodic low-conductor part}}_{\text{transferred here}} + \underbrace{\text{energy-bearing nonperiodic residual}}_{\text{uncontrolled at fixed } h}. \quad (63)$$

Moving from (63) toward a fixed prime-pair problem requires at least one genuinely new ingredient. Possible stages are:

- (1) replace the residue-class triangle inequality by an L^2 or coefficient-sensitive estimate and determine the largest rigorously transferable periodic complexity;
- (2) construct a structured nonperiodic coefficient class whose residual has provably smaller bilinear influence, not merely smaller local Fourier data;

- (3) obtain Type II or dispersion control for the energy-bearing residual against $\Lambda(mn + h)$ at a prescribed h ;
- (4) only after such control, reconnect the exact divisor ledger to a fixed prime-pair correlation.

Stages (2) and (3) are not supplied by classical Bombieri–Vinogradov. They are the mathematical gap, rather than an omitted routine estimate. A negative result showing that a proposed coefficient class cannot reduce (60) would also be a legitimate outcome of the next stage; it would narrow the route without implying a universal parity impossibility theorem.

6 Exact certificate, numerical diagnostic, and conclusion

The proofs above are analytic and do not depend on computation. The accompanying package serves two more limited purposes: it checks every finite normalization entering the operator, and it makes the difference between an exact arithmetic identity and a finite Mangoldt experiment visible.

6.1 The integer/rational certificate

The reference certificate uses

$$q = 5 \cdot 7 \cdot 11 = 385, \quad h = 2. \quad (64)$$

Here $\varphi(q) = 240$,

$$\kappa_q = \frac{9}{16}, \quad R_q = \varphi(q)\kappa_q = 135. \quad (65)$$

The program `experiments/exact_transfer_certificate.py` uses only integers and exact rational arithmetic from the standard library. It verifies the following items.

- (i) The pointwise inclusion-exclusion identity is checked for all $240^2 = 57,600$ unit pairs.
- (ii) For every $d \mid 385$, the map $J_{2,d}(a) = -2a^{-1}$ is checked to be an involution. Its fixed points, two-cycles, and exact ± 1 eigenspace dimensions are recorded.
- (iii) The direct vector $P_{385,2}\beta$ is compared entry by entry with the involution-bundle formula (13) for a deterministic signed observable.
- (iv) The principal mode and the seven nonprincipal real quadratic modes of exact conductor $d \mid 385$ are checked. Their normalized multiplier magnitudes are exactly

$$1, \frac{1}{3}, \frac{1}{5}, \frac{1}{9}, \frac{1}{15}, \frac{1}{27}, \frac{1}{45}, \frac{1}{135}, \quad (66)$$

corresponding respectively to $d = 1, 5, 7, 11, 35, 55, 77, 385$.

- (v) The row-main identity

$$\frac{mN}{\varphi(mq)} \sum_{\substack{b \in G_q \\ (mb+h, q)=1}} \beta(b) = \frac{mN}{\varphi(m)} (P_{q,h}\beta)(m) \quad (67)$$

is verified exactly on every eligible test row.

- (vi) The reindexing (32) is checked independently for an arbitrary integer target weight. It is also checked for a formal von Mangoldt function represented as an integer coefficient vector in the independent basis $\{\log p\}$; no floating logarithm is used.

The canonical payload SHA-256 is

030F20B495E68234CD3176795C1F3652D2F8792DE88C1BE49BC6A04F15FBCCFF.

The committed JSON file SHA-256 is

2EEE645F124CCDA3A50F4FB349550AC7195321EA5FC4410853142CD655E1E37E.

Regeneration is byte deterministic. These hashes certify finite data and code conventions, not the Bombieri–Vinogradov estimate.

6.2 A separate finite Mangoldt diagnostic

The optional NumPy script `experiments/lambda_transfer_diagnostics.py` sieves the genuine von Mangoldt function and computes each row twice: directly in n , and by target arithmetic progressions modulo mq . It then compares the common row value with the exact rational local main term. This is a sharp-box diagnostic on $N < n \leq 2N$, corresponding formally to $\mathcal{I}_W = 1$, rather than an implementation of the smooth weight in the theorem. A reference run takes

$$M = 32, \quad N = 2^{19}, \quad X = MN = 2^{24}, \quad q = 35, \quad h = 2. \quad (68)$$

There are 11 eligible rows. The reported quantities are shown in [table 1](#). The conductor- d observables are the real quadratic modes used by the script.

Table 1: Finite Mangoldt row diagnostic at (68).

d	$\sum_m \text{main}_m /X$	$\sum_m \text{error}_m /X$	RMS/ N	max/ N
1	2.6569×10^{-1}	7.1598×10^{-4}	2.4501×10^{-3}	4.4714×10^{-3}
5	8.8565×10^{-2}	6.9698×10^{-4}	3.0493×10^{-3}	8.0454×10^{-3}
7	5.3139×10^{-2}	5.4323×10^{-4}	2.2140×10^{-3}	5.3955×10^{-3}
35	1.7713×10^{-2}	8.6667×10^{-4}	2.8950×10^{-3}	4.7501×10^{-3}

The largest discrepancy between the direct and progression computations is 1.17×10^{-10} , at floating precision. The second numeric column is the finite supremum of the displayed row error over coefficients $|\alpha_m| \leq 1$, normalized by X .

This experiment is not a verification of [theorem 3.1](#). At the reference scale, $2Mq/\sqrt{X} = 0.546875$; the non-explicit logarithmic constant B in the asymptotic theorem is not instantiated. Nor does a small error for four finite modes say anything about the energy-bearing residual in (60). The floating output is therefore intentionally separate from the canonical exact certificate.

6.3 Reproducibility commands

From the paper directory, the exact artifact and its zero-dependency test runner are produced first; the floating smoke test additionally requires NumPy:

```
python experiments/exact_transfer_certificate.py \
  --output experiments/data/exact-certificate.json
python experiments/test_exact_transfer_certificate.py
python experiments/test_lambda_transfer_diagnostics.py
```

The test modules are also discoverable by `pytest`. The floating reference run is

```
python experiments/lambda_transfer_diagnostics.py \
  --m 32 --n 524288 --q 35 --h 2 \
  --conductors 1,5,7,35 \
  --output experiments/data/lambda-diagnostic.json
```

Its output is not part of the exact certificate.

6.4 Conclusion

The paper advances the fixed-shift program by one controlled layer. An arbitrary second coefficient remains outside the one-sided Bombieri–Vinogradov reduction, but a bounded low-period coefficient can be passed through $\Lambda(mn + h)$ with an ℓ^1 row error. The leading term is not merely a density: it is the exact finite operator $P_{q,h}$. Its divisor-bundle involutions, conjugate-mode exchange, higher conductor-exponent kernel, damping factors, and even trace moments are all explicit.

The same result also draws a sharper boundary. The actual modulus is mq , the residue complexity costs q , and TPC-7 shows that the permitted polylogarithmic periodic spaces miss asymptotically all of the $\Lambda - 1$ energy. Thus the theorem genuinely reaches a prime target at a prescribed h , including $h = 2$, while stopping before the Type II residual that contains the fixed prime-pair difficulty. That separation is the main conclusion: a finite arithmetic dynamics survives the prime target exactly, but it is not a substitute for the missing parity-sensitive estimate.

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